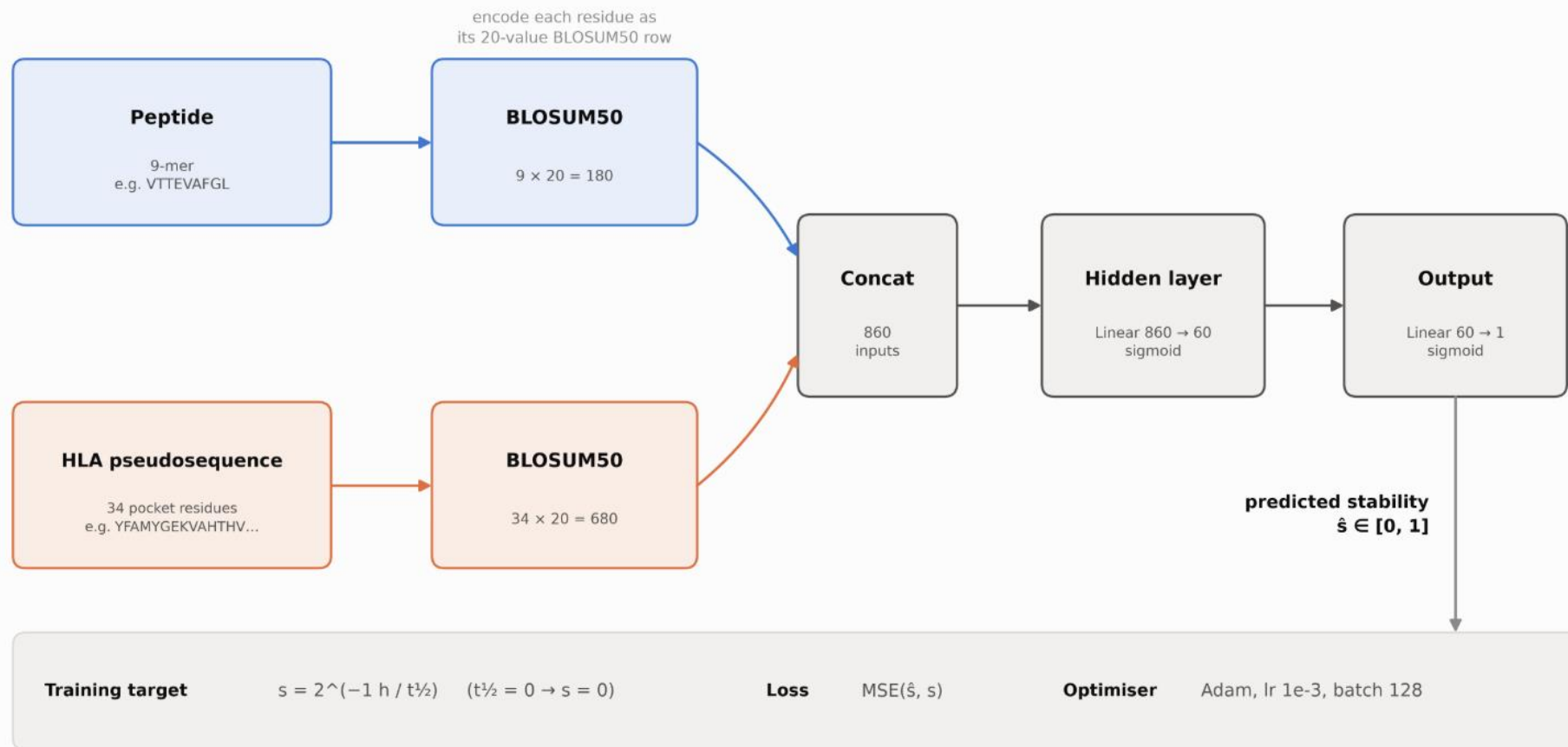


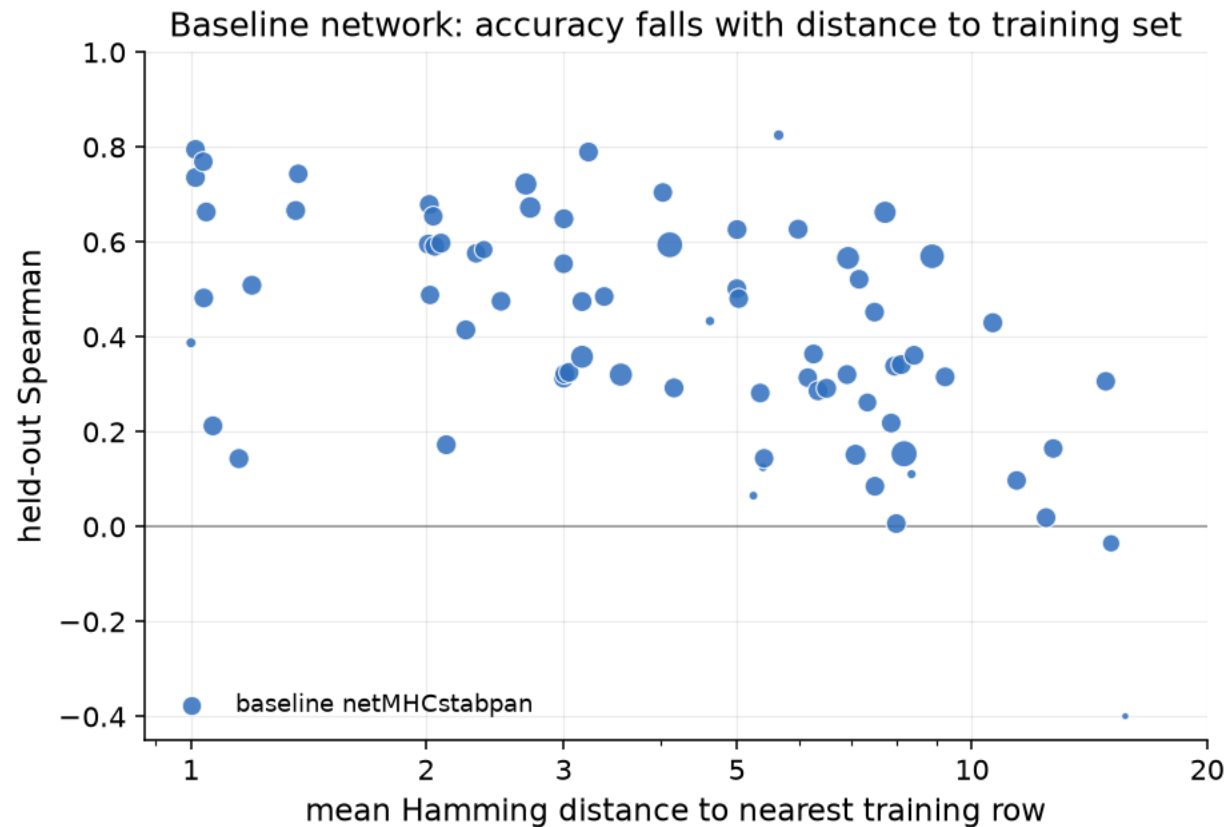
Baseline Model Architecture

Single hidden layer, sigmoid units · after NetMHCstabpan (Rasmussen et al. 2016) · 51,721 trainable parameters



Baseline Model Overfitting

- Train spearman is ~ 0.85 and test spearman is ~ 0.45 .



Trying to reduce overfitting

- Reducing input size:
 - Using only P2 and P9 in the peptide.
 - Using only 20 relevant positions in the HLA sequence from pocket B and F.
 - Compressing dimension of the BLOSUM-50 matrix.
- Early stopping
- Dropout